TECHNICAL DESIGN DOCUMENT

Pharmacovigilance Intelligence Platform

From Raw Adverse Event Reports to AI-Powered Safety Signals

Data Engineering Capstone Project DAAI_C2C_2026

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Technology Stack

AWS S3 | Dataiku DSS | Snowflake Data Cloud | Snowflake Cortex Forecast | SQL | Microsoft Power BI

1. Project Overview

1.1 Introduction

Pharmacovigilance is the process of monitoring, assessing, and preventing adverse effects associated with pharmaceutical products to ensure patient safety. Meridian Therapeutics receives adverse event (AE) reports from healthcare professionals, patients, and regulatory agencies across multiple countries. These reports are critical for identifying potential safety risks; however, the existing process relies on manually reviewing raw CSV exports from a legacy safety system, making it time-consuming, error-prone, and difficult to detect emerging safety signals promptly.

To address these challenges, a cloud-native **Pharmacovigilance Intelligence Platform** was developed using AWS, Dataiku DSS, Snowflake, and Power BI. The platform automates data ingestion, profiling, cleansing, dimensional modeling, AI/ML-based forecasting, and interactive reporting, transforming raw adverse event data into trusted, analytics-ready information.

1.2 Project Objectives

The solution was designed to achieve the following objectives:

- Automate ingestion of adverse event reports into a secure cloud environment.
- Improve data quality through profiling, cleansing, standardization, and validation.
- Design a dimensional data model optimized for analytical reporting.
- Implement Snowflake analytical views for efficient querying.
- Apply Snowflake AI/ML forecasting to support early identification of abnormal reporting trends.
- Deliver an interactive Power BI dashboard for monitoring adverse event patterns across drugs, manufacturing batches, countries, and reporting periods.

1.3 Solution Summary

The implemented solution follows a modern data engineering architecture. Raw adverse event reports are uploaded to AWS S3, ingested into the Snowflake RAW layer, processed in Dataiku DSS for data quality management, and then loaded into the Snowflake STAGING and CURATED layers for analytics, AI/ML forecasting, and reporting. Dataiku DSS performs data profiling, cleansing, standardization, and deduplication before the validated dataset is promoted to the Snowflake STAGING layer. A curated star schema and analytical views are then created to support reporting and AI/ML forecasting. Finally, Power BI provides interactive dashboards that enable safety reviewers to monitor adverse event trends and forecast future reporting volumes for proactive decision-making.

2. Solution Architecture

2.1 Architecture Overview

The Pharmacovigilance Intelligence Platform follows a layered cloud-native data engineering architecture that separates data ingestion, processing, storage, analytics, and visualization into independent components. This modular design improves scalability, maintainability, and data governance while enabling each stage of the pipeline to perform a clearly defined function.

Raw adverse event reports are ingested into AWS S3, where they are securely stored before being loaded into the Snowflake **RAW** layer. Dataiku DSS performs data profiling, cleansing, standardization, validation, and deduplication to produce high-quality datasets, which are then promoted to the Snowflake **STAGING** layer. Curated fact and dimension tables, along with analytical views, are created in the CURATED layer to support business reporting, AI/ML forecasting, and downstream analytics. Snowflake AI/ML forecasting models analyze adverse event trends, and the processed data is finally presented through interactive Power BI dashboards for business users.

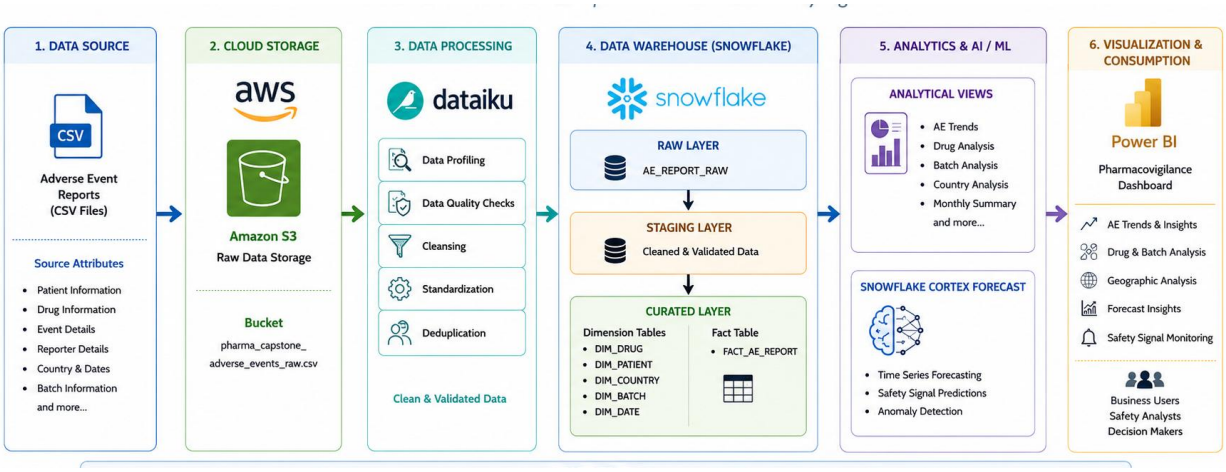


Figure 2.1 illustrates the end-to-end architecture of the implemented solution.

2.2 Architecture Components

Component	Purpose
AWS S3	Securely stores raw adverse event report files and acts as the landing zone for data ingestion.
Snowflake - RAW	Preserves source data in its original format, ensuring traceability and auditability.
Dataiku DSS	Profiles, cleanses, standardizes, validates, and deduplicates data through an auditable ETL workflow.

Component	Purpose
Snowflake – STAGING	Stores validated datasets prepared for downstream transformation and dimensional modeling.
Snowflake – CURATED	Contains the dimensional data model, analytical views, and optimized datasets for reporting and AI/ML processing.
Snowflake Cortex Forecast	Predicts future adverse event reporting volumes using historical time-series data to support proactive safety monitoring.
Power BI	Delivers interactive dashboards and KPIs for monitoring adverse event trends and supporting data-driven decision-making.

3. End-to-End Data Engineering Pipeline

Pipeline Stages

Stage	Description
1. Data Ingestion	Raw adverse event report files are uploaded to AWS S3, providing a secure and centralized landing zone for source data.
2. Raw Data Storage	Source files are loaded into the Snowflake **RAW** schema without modification, preserving the original data for traceability and audit purposes.
3. Data Profiling & Cleansing	Dataiku DSS profiles the dataset to identify data quality issues, followed by cleansing, standardization, validation, and deduplication through an auditable ETL workflow.
4. Staging & Data Modeling	The validated dataset is loaded into the Snowflake **STAGING** schema and transformed into a dimensional model consisting of fact and dimension tables within the **CURATED** schema.
5. AI/ML Processing	Snowflake Cortex Forecast analyzes historical adverse event reporting patterns to predict future reporting volumes, supporting proactive safety monitoring.
6. Business Intelligence	Curated datasets and forecast outputs are consumed by Power BI dashboards, enabling interactive analysis of adverse event trends, manufacturing batches, countries, and reporting periods.

4. Data Ingestion and Storage

4.1 Data Ingestion Overview

The data ingestion layer is the first stage of the Pharmacovigilance Intelligence Platform and is responsible for securely capturing raw adverse event (AE) reports from the source environment and making them available for downstream processing.

The platform follows a **batch ingestion approach**, where adverse event reports generated from the source safety system are provided as CSV files. These files are initially stored in Amazon Simple Storage Service (AWS S3), which acts as the centralized cloud landing zone for raw data.The ingestion design separates data acquisition from data processing by preserving the original source files before any transformation is applied. This approach ensures that the original data remains available for validation, auditing, and reprocessing whenever required.

The overall ingestion flow is:

Source CSV Files → AWS S3 Landing Zone → Snowflake RAW Layer → Data Quality Processing

This layered ingestion approach improves data governance, provides traceability, and establishes a reliable foundation for analytics and machine learning workloads.

4.2 AWS S3 Landing Zone Implementation

Amazon S3 was selected as the initial storage layer due to its high durability, scalability, cost-effectiveness, and seamless integration with Snowflake.The S3 bucket functions as a **raw data repository**, where incoming adverse event reports are stored in their original format. No cleansing, filtering, or transformation is performed at this stage.

Maintaining the original source file is important in pharmacovigilance systems because regulatory and safety analytics workflows require the ability to trace processed information back to the original reported data.

The uploaded source dataset used in this implementation was:

Property	Description
File Name	pharma_capstone_adverse_events_raw.csv
File Format	CSV
Data Domain	Adverse Event Reports
Storage Layer	AWS S3 Landing Zone
Processing Status	Raw / Unprocessed

The S3 landing zone provides the following capabilities:

- Centralized storage location for incoming safety data.
- Preservation of source data before transformation.
- Support for repeatable ingestion processes.
- Separation between raw data storage and analytical processing.

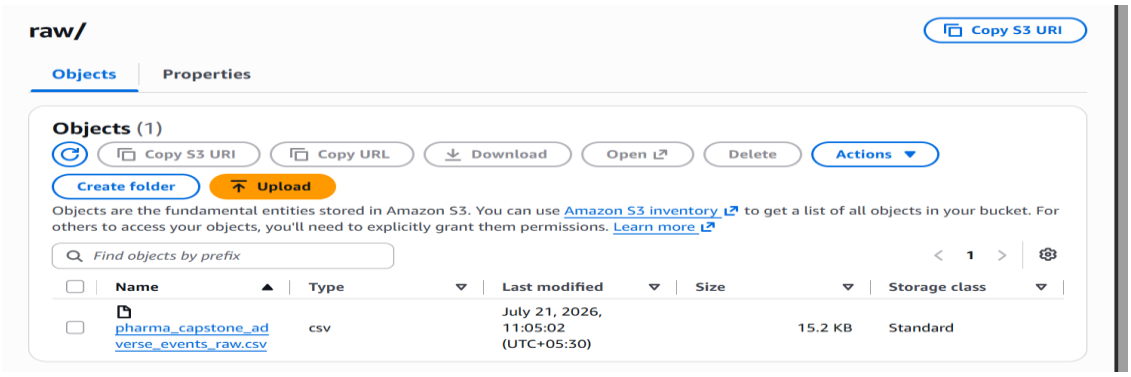


Figure 4.1 shows the raw adverse event dataset stored in the AWS S3 landing zone.

4.3 Snowflake RAW Layer Implementation

After ingestion into AWS S3, the raw adverse event dataset was loaded into the Snowflake RAW layer.The RAW layer serves as the landing layer in the implemented medallion architecture, preserving source data in its original form before downstream processing, where source data is first stored in an unchanged format before moving through subsequent processing layers.

The purpose of this layer is not to provide analytical insights but to maintain a trusted copy of the ingested source data. All data quality operations, including profiling, standardization, and deduplication, are performed after the data is loaded into this layer.

The Snowflake RAW layer was implemented using the following structure:

Snowflake Object	**Implementation**
Database	PHARMA_DB
Schema	RAW
Table	AE_REPORT_RAW

The AE_REPORT_RAW table contains the original adverse event attributes received from the source file, including:

- Report identifiers
- Report dates
- Patient demographics
- Drug information
- Adverse event details
- Manufacturing information

The source column structure was preserved during ingestion to maintain data lineage and ensure that downstream transformations could be validated against the original records.

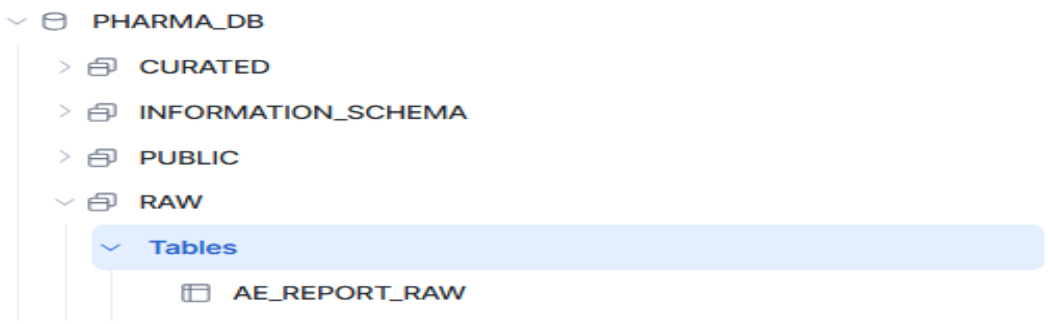


Figure 4.3.1: RAW Layer Implementation in Snowflake (PHARMA_DB → RAW → AE_REPORT_RAW)

4.4 Raw Data Loading Process

The ingestion process implemented in the platform follows the below sequence:

Step 1: Source Data Availability

The adverse event report file is generated from the source safety system and provided in CSV format.

Step 2: Cloud Storage Ingestion

The file is uploaded into AWS S3, which acts as the secure landing location for raw data.

Step 3: Snowflake Loading

The raw dataset is loaded into the Snowflake RAW schema, where it is stored without applying business transformations.

Step 4: Data Quality Processing

The RAW dataset becomes the input source for Dataiku DSS, where profiling, cleansing, validation, and deduplication activities are performed.

This separation ensures that data ingestion and data transformation responsibilities remain independent, improving maintainability and governance.

5. Data Quality Management Using Dataiku DSS

5.1 Data Quality Processing Overview

Data quality management is a critical stage in the Pharmacovigilance Intelligence Platform because adverse event reports received from operational systems may contain incomplete, inconsistent, or duplicate information. Poor data quality can directly impact safety analysis, reporting accuracy, and downstream machine learning results.

To address these challenges, Dataiku DSS was used as the data preparation and quality management platform. The Dataiku workflow was designed to profile the incoming dataset, identify quality issues, apply required transformations, and generate a validated dataset suitable for analytical processing.

The Dataiku pipeline follows a structured data preparation workflow:

Snowflake RAW Dataset → Data Profiling → Data Cleansing → Standardization → Validation → Deduplication → Clean Analytical Dataset

This approach ensures that only validated and standardized data is promoted to downstream Snowflake layers.

5.2 Data Profiling and Quality Assessment

Before applying transformations, the raw adverse event dataset was analyzed using Dataiku DSS profiling capabilities.

The profiling stage provided visibility into:

- Dataset structure and schema
- Column-level statistics
- Missing value patterns
- Data inconsistencies
- Duplicate records
- Value distributions

The profiling results were used to identify data quality issues that required remediation before analytical processing.

The initial dataset characteristics were:

Attribute	Value
Dataset	AE_REPORT_RAW_DS
Initial Record Count	83
Columns	19

Attribute	Value
Data Format	CSV
Processing Tool	Dataiku DSS

5.3 Identified Data Quality Issues

The profiling process identified several data quality challenges within the adverse event dataset.

Missing and Unknown Values

Certain attributes contained incomplete information, particularly patient demographic fields.

Example:

Column	Issue Identified
PATIENT_AGE	Multiple null and unknown values
PATIENT_SEX	Missing values and inconsistent representations

These issues required standardization to ensure consistent analytical interpretation.

Inconsistent Data Values

Some categorical fields contained different representations of the same value.

Example:

Before standardization:

- Male
- male
- M
- MALE

After standardization:

- MALE
- FEMALE
- UNKNOWN

Standardizing categorical attributes improves filtering, aggregation, and reporting accuracy.

Date Format Inconsistencies

The dataset contained multiple date formats, which could affect time-based analysis and forecasting.Dataiku date parsing was applied to convert different date representations into a consistent date format suitable for Snowflake analytical processing.

Duplicate Records

Duplicate analysis identified repeated adverse event records based on report identifiers.Duplicate records were reviewed and removed to ensure that each adverse event report was represented only once in the analytical dataset.

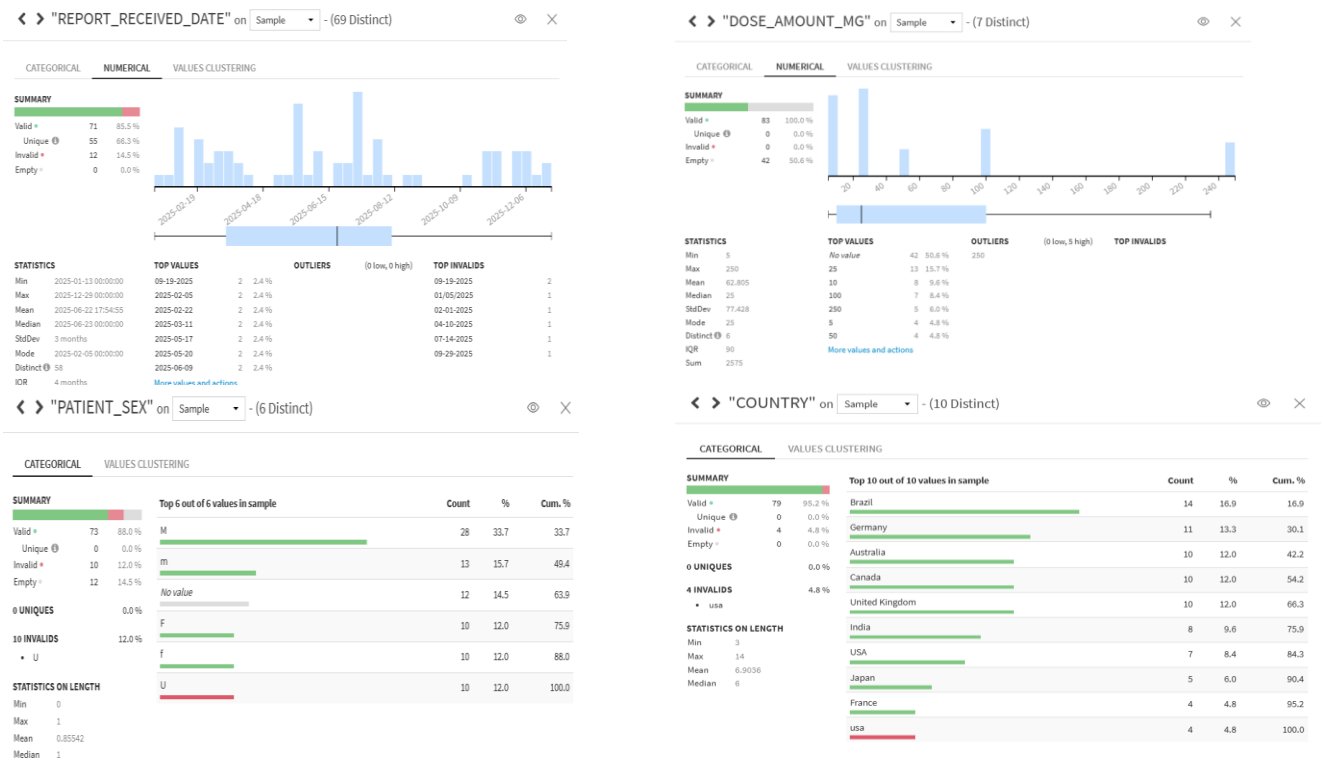


Figure 5.1 illustrates the Dataiku profiling results and identified data quality issues.

5.4 Data Cleansing and Transformation Workflow

After identifying data quality issues, a series of Dataiku transformation steps were applied to improve dataset reliability.

The main cleansing activities included:

Column Standardization

Column names and categorical values were standardized to maintain consistency across the analytical platform.

Examples:

- Converted text values into uppercase format.
- Standardized categorical attributes.
- Removed inconsistent formatting.

Missing Value Handling

Missing and unknown values were analyzed and standardized.

Instead of removing incomplete records, appropriate handling strategies were applied to preserve valuable adverse event information while maintaining analytical consistency.

Date Normalization

Dataiku date parsing functions were used to convert different source date formats into a unified date representation.

This ensured compatibility with:

- Snowflake time-series analysis
- Monthly aggregation
- Forecasting models
- Power BI reporting

Duplicate Detection and Removal

Duplicate handling was implemented through Dataiku grouping and window-based processing.

The workflow identified duplicate REPORT_ID values and retained the appropriate unique records for downstream analysis.

5.5 Data Quality Validation

After completing the cleansing workflow, the processed dataset was validated to confirm that transformation rules were correctly applied.

The final cleaned dataset characteristics were:

Attribute	Value
Final Dataset	AE_REPORT_CLEAN_DS_Final
Final Record Count	81
Columns	23
Duplicate Records	Removed
Data Status	Validated and Analytics - Ready

Validation checks performed included:

- Record count verification
- Duplicate verification
- Data type validation
- Standardized category validation
- Date format validation

The validated dataset was then prepared for loading into the Snowflake STAGING layer for dimensional modeling and analytical processing.

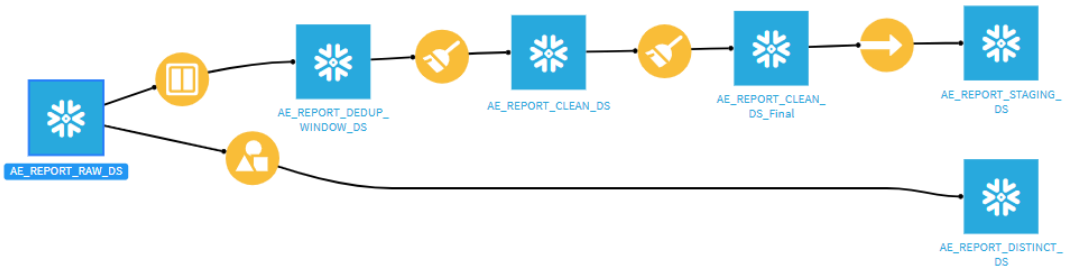


Figure 5.2 illustrates the Dataiku preparation flow used for cleansing and transformation.

6. Snowflake Data Modeling and Curated Layer Design

6.1 Data Modeling Overview

After completing data quality processing in Dataiku DSS, the validated adverse event dataset was loaded into the Snowflake STAGING layer for further transformation and analytical modeling.

A dimensional data modeling approach was implemented within the Snowflake CURATED layer to organize pharmacovigilance data into a structure optimized for reporting, analytical queries, and machine learning workloads.

The curated model follows a **star schema design**, where a central fact table stores measurable adverse event information and surrounding dimension tables store descriptive business attributes.

The overall data flow is:

STAGING Layer → Dimensional Transformation → CURATED Star Schema → Analytics / AI-ML / Reporting

This design provides a scalable and maintainable structure for analyzing adverse event patterns across multiple business dimensions such as drugs, manufacturing locations, reporting periods, and geographical information.

6.2 Star Schema Design Approach

A star schema was selected because it provides an efficient structure for analytical workloads.

The design separates:

Fact Data

Contains measurable events and numerical values used for analysis.

Example:

- Number of adverse event reports.
- Reporting frequency.
- Time-based event measurements.

Dimension Data

Contains descriptive attributes that provide context for analysis.

Example:

- Drug information.
- Manufacturing site details.
- Reporting dates.

This approach improves:

- Query performance.
- Data organization.

- Dashboard development.
- Analytical flexibility.

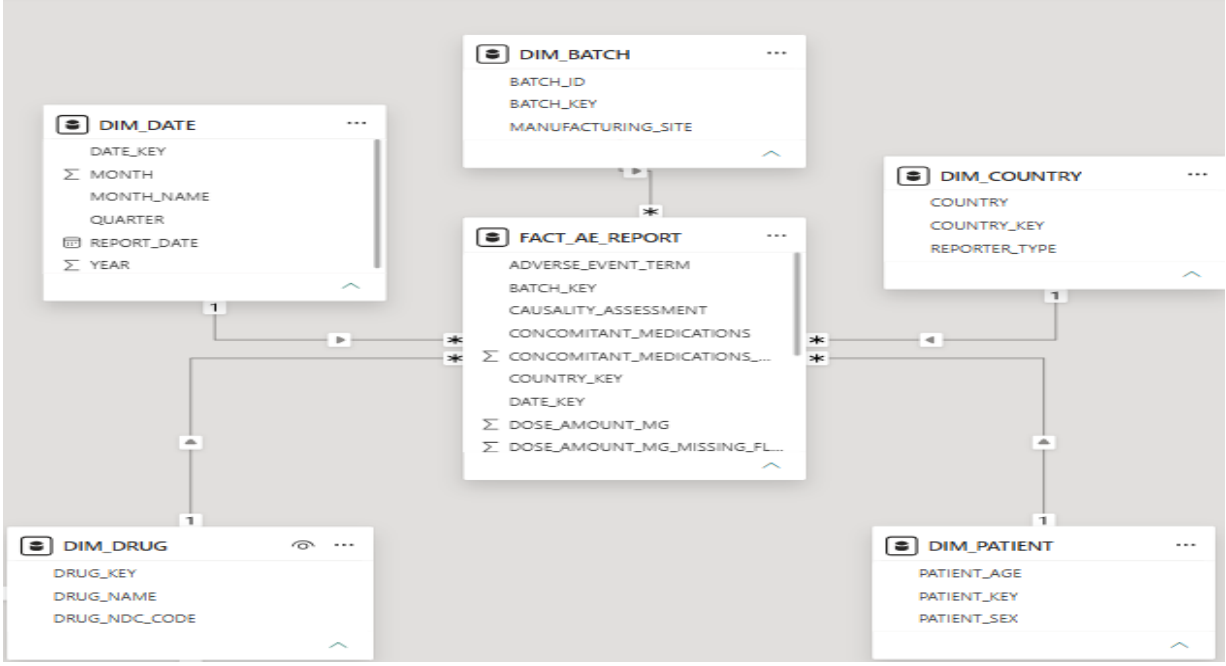


Figure 6.2.1: Pharmacovigilance Star Schema Design Implemented in Snowflake CURATED Layer

6.3 Curated Data Model Structure

The implemented star schema consists of a central adverse event fact table connected with supporting dimension tables.

Fact Table

FACT_AE_REPORT

The fact table stores adverse event reporting measurements used for analysis.

Attribute	Description
AE_REPORT_ID	Unique adverse event identifier
DRUG_KEY	Reference to drug dimension
SITE_KEY	Reference to manufacturing site dimension
DATE_KEY	Reference to date dimension
AE_REPORT_COUNT	Number of adverse event reports

The fact table enables aggregation of adverse event volumes across different analytical dimensions.

Dimension Tables

DIM_DRUG

Stores drug-related descriptive information.

Purpose:

- Analyze adverse event patterns by medication.
- Compare safety trends across different products.

Example attributes:

Attribute	Description
DRUG_KEY	Unique drug identifier
DRUG_NAME	Drug product name
DRUG_NDC_CODE	Product identification code

DIM_MANUFACTURING_SITE

Stores manufacturing location information.

Purpose:

- Identify reporting patterns by production facility.
- Support manufacturing-level safety monitoring.

Example attributes:

Attribute	Description
SITE_KEY	Unique site identifier
MANUFACTURING_SITE	Production facility name

DIM_DATE

Stores calendar attributes required for time-based analysis.

Purpose:

- Enable monthly, quarterly, and yearly trend analysis.
- Support forecasting model requirements.

Example attributes:

Attribute	Description
DATE_KEY	Unique date identifier
REPORT_DATE	Adverse event reporting date
REPORT_MONTH	Monthly aggregation period
YEAR	Reporting year
MONTH	Reporting month

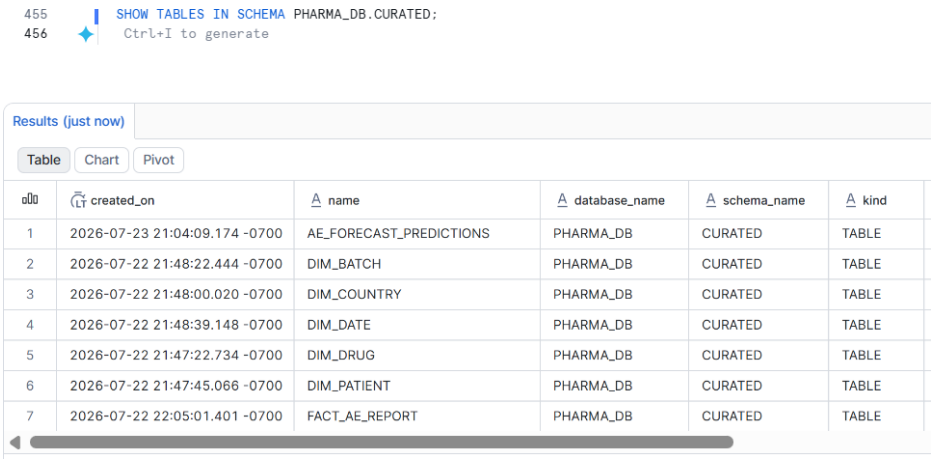


Figure 6.3.1: CURATED Schema Structure Showing Fact and Dimension Tables in Snowflake

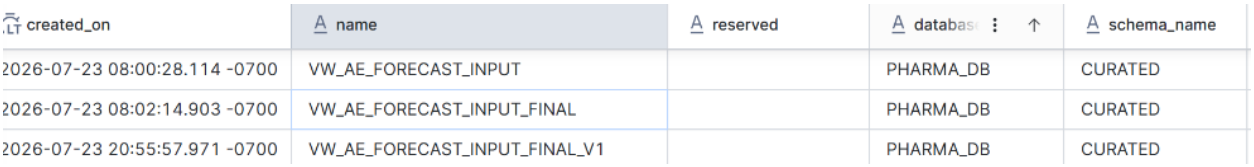
6.4 Analytical Views Implementation

In addition to physical tables, analytical views were created in the CURATED layer to support reporting and machine learning requirements.

These views provide pre-processed datasets without requiring repeated transformation logic.

Implemented analytical views include:

View Name	Purpose
VW_AE_DRUG_BATCH_MONTHLY	Aggregates adverse event counts by drug, batch, manufacturing site, and month
VW_AE_ML_INPUT	Provides prepared input data for machine learning workloads
VW_AE_FORECAST_INPUT_FINAL	Formats historical data for Snowflake forecasting



created_on	name	reserved	database	schema_name
2026-07-23 08:00:28.114 -0700	VW_AE_FORECAST_INPUT		PHARMA_DB	CURATED
2026-07-23 08:02:14.903 -0700	VW_AE_FORECAST_INPUT_FINAL		PHARMA_DB	CURATED
2026-07-23 20:55:57.971 -0700	VW_AE_FORECAST_INPUT_FINAL_V1		PHARMA_DB	CURATED

Figure 6.4.1: Analytical Views Created in Snowflake CURATED Layer for Reporting and ML Workloads

7. Analytical Views and Data Preparation for AI/ML

7.1 Analytical Data Preparation Overview

After implementing the Snowflake CURATED layer, analytical views were created to simplify data access for reporting and machine learning workloads.

While the dimensional model provides a structured representation of pharmacovigilance data, analytical workloads often require aggregated and purpose-specific datasets. To address this requirement, dedicated Snowflake views were developed to provide optimized datasets for different consumption scenarios.

The analytical preparation layer performs the following functions:

- Aggregates adverse event reporting metrics.
- Combines relevant business attributes.
- Prepares time-series datasets.
- Creates structured inputs for Snowflake Cortex forecasting models.
- Provides reusable datasets for Power BI reporting.

The analytical workflow follows:

CURATED Tables → Analytical Views → BI Reporting / AI-ML Models

7.2 Adverse Event Trend Aggregation View

VW_AE_DRUG_BATCH_MONTHLY

The VW_AE_DRUG_BATCH_MONTHLY view was created to analyze adverse event reporting trends across different drugs, manufacturing batches, and reporting periods.

The view aggregates individual adverse event records into monthly reporting metrics, allowing safety teams to identify changes in reporting frequency over time.

The aggregation was performed using key business attributes:

- Drug Name
- Batch ID
- Manufacturing Site
- Reporting Year
- Reporting Month

The resulting dataset provides monthly adverse event counts that can be used for:

- Trend analysis.
- Drug safety monitoring.
- Manufacturing-level analysis.
- Machine learning preparation.

Example output structure:

Column	Description
DRUG_NAME	Pharmaceutical product name
BATCH_ID	Manufacturing batch identifier
MANUFACTURING_SITE	Production location
REPORT_MONTH	Monthly reporting period
AE_REPORT_COUNT	Number of adverse event reports

	DRUG_NAME	BATCH_ID	MANUFACTURING_SITE	REPORT_DATE	AE_REPORT_COUNT
1	GLUCOMEND	B1030-A	MFG-DE-BER-02	2025-01-22	1
2	PULMOCARE-XR	B1000-B	MFG-IN-HYD-03	2025-02-02	1
3	DERMAFEX	B1066-C	MFG-IN-HYD-03	2025-02-05	1
4	VASCULIN	B1036-B	MFG-BR-SAO-04	2025-08-25	1
5	CARDIOZAN	B1059-C	MFG-IN-HYD-03	2025-07-05	1
6	CARDIOZAN	B1043-C	MFG-DE-BER-02	2025-07-11	1
7	NEUROLEX	B1091-B	MFG-DE-BER-02	2025-12-12	1
8	VASCULIN	B1080-B	MFG-JP-OSA-05	2025-05-17	1
9	IMMUNOVA	B1085-B	MFG-JP-OSA-05	2025-09-19	1
10	VASCULIN	B1058-B	MFG-BR-SAO-04	2025-11-06	1

Figure 7.2.1: VW_AE_DRUG_BATCH_MONTHLY View Output for Adverse Event Trend Analysis

7.3 Machine Learning Input Dataset Preparation

Snowflake AI/ML models require structured and consistent input data. Therefore, additional views were created to prepare historical adverse event reporting data for forecasting.

VW_AE_ML_INPUT

This view provides the initial machine learning dataset by selecting required attributes from the aggregated analytical data.

The dataset includes:

- Drug information.
- Manufacturing information.
- Reporting period.
- Historical adverse event counts.

Purpose:

- Provides a reusable ML input layer.
- Separates ML preparation logic from business tables.
- Ensures consistent input structure for model training.

	DRUG_NAME	BATCH_ID	MANUFACTURING_SITE	REPORT_DATE	AE_REPORT_COUNT
1	NEUROLEX	B1064-C	MFG-US-ATL-01	2025-01-13	1
2	RHEUMATRIX	B1009-C	MFG-US-ATL-01	2025-02-04	1
3	IMMUNOVA	B1005-A	MFG-BR-SAO-04	2025-11-30	1
4	VASCULIN	B1043-A	MFG-JP-OSA-05	2025-11-07	1
5	RHEUMATRIX	B1040-B	MFG-IN-HYD-03	2025-06-28	1
6	DERMAFEX	B1020-B	MFG-JP-OSA-05	2025-02-07	1
7	CARDIOZAN	B1017-A	MFG-US-ATL-01	2025-11-23	1
8	IMMUNOVA	B1085-B	MFG-JP-OSA-05	2025-09-19	1
9	GLUCOMEND	B1090-B	MFG-BR-SAO-04	2025-06-09	1
10	ONCOZEL	B1088-B	MFG-BR-SAO-04	2025-12-08	1

Figure 7.3.1: VW_AE_ML_INPUT View Structure for Snowflake Machine Learning Preparation

7.4 Forecasting Dataset Preparation

VW_AE_FORECAST_INPUT_FINAL

The forecasting model requires a time-series dataset with:

- A timestamp column.
- A series identifier.
- A target numerical value.

The forecasting input view was created to satisfy these requirements.

The dataset was prepared with the following components:

Forecast Requirement	Implementation
Time Column	REPORT_MONTH
Series Identifier	MANUFACTURING_SITE
Target Variable	AE_REPORT_COUNT

The prepared dataset represents historical adverse event reporting volumes by manufacturing site over time.

This structure enables Snowflake Cortex Forecast to learn historical reporting patterns and generate future adverse event reporting predictions.

	MANUFACTURING_SITE	REPORT_MONTH	AE_REPORT_COUNT
1	MFG-US-ATL-01	2025-01-01 00:00:00.000	1
2	MFG-BR-SAO-04	2025-01-01 00:00:00.000	1
3	MFG-JP-OSA-05	2025-01-01 00:00:00.000	0
4	MFG-DE-BER-02	2025-01-01 00:00:00.000	1
5	MFG-IN-HYD-03	2025-01-01 00:00:00.000	0
6	MFG-US-ATL-01	2025-02-01 00:00:00.000	5
7	MFG-BR-SAO-04	2025-02-01 00:00:00.000	0
8	MFG-JP-OSA-05	2025-02-01 00:00:00.000	2
9	MFG-DE-BER-02	2025-02-01 00:00:00.000	1
10	MFG-IN-HYD-03	2025-02-01 00:00:00.000	3

Figure 7.4.1: VW_AE_FORECAST_INPUT_FINAL View Prepared for Snowflake Cortex Forecasting

7.5 Forecasting Input Design Decision

The forecasting model was designed using **MANUFACTURING_SITE as the forecasting series dimension**.

This approach was selected because manufacturing locations represent operational units where changes in adverse event reporting trends may provide valuable safety monitoring insights.

By forecasting reporting volumes at the manufacturing site level, the platform supports:

- Monitoring changes in reporting behavior.
- Identifying increasing reporting patterns.
- Supporting proactive safety review.

8. Snowflake AI/ML Forecasting

8.1 Forecasting Overview

To enhance the analytical capabilities of the Pharmacovigilance Intelligence Platform, Snowflake Cortex Forecasting was integrated into the solution to predict future adverse event reporting volumes. Rather than relying solely on historical analysis, forecasting enables the

platform to estimate future reporting patterns, allowing safety teams to monitor potential increases in adverse event activity and plan investigations proactively.

The forecasting process was implemented using historical adverse event reporting data prepared within the CURATED layer. Aggregated monthly reporting counts were used as the input for model training, ensuring that the forecasting model learned from validated and analytics-ready data.

The overall forecasting workflow is illustrated below:

Historical Curated Data → Forecast Input View → Snowflake Cortex Forecasting Model → Future Reporting Predictions

This approach extends the platform from descriptive analytics to predictive analytics, supporting more informed decision-making in pharmacovigilance operations.

8.2 Forecasting Model Implementation

A forecasting model was created using Snowflake Cortex Forecasting to estimate future adverse event report volumes.The model was trained using historical monthly reporting data grouped by manufacturing site. Each manufacturing site was treated as an independent time series, allowing the model to learn reporting patterns specific to each production location.

The forecasting dataset contained:

- Historical reporting period
- Manufacturing site identifier
- Monthly adverse event report count

The prepared dataset was supplied to the forecasting model after validating the timestamp format and aggregation logic.

8.3 Forecasting Model Configuration

The forecasting model was configured with the following parameters:

Configuration Parameter	Value
Forecasting Engine	Snowflake Cortex Forecasting
Input Dataset	VW_AE_FORECAST_INPUT_FINAL
Time Column	REPORT_MONTH_v1
Series Column	MANUFACTURING_SITE
Target Variable	AE_REPORT_COUNT

Configuration Parameter	Value
Forecast Horizon	3 Months
Output	Forecasted Monthly AE Report Counts

The selected configuration enables independent forecasting for each manufacturing site while preserving historical reporting trends.

8.4 Forecast Generation Process

Once the forecasting model was successfully trained, prediction requests were executed to generate future adverse event reporting volumes.

The forecasting process consisted of the following steps:

1. Historical adverse event data was aggregated at the monthly level.
2. The forecasting input view was prepared with the required timestamp, series, and target columns.
3. Snowflake Cortex Forecasting trained the model using historical observations.
4. Forecasts were generated for the specified prediction horizon.
5. Predicted results were stored in a Snowflake table for downstream reporting and visualization.

The generated predictions included not only the expected forecast values but also confidence intervals, providing an estimated range within which future reporting volumes are likely to occur.

8.5 Forecast Results

The forecasting model produced predicted adverse event reporting volumes for future reporting periods based on historical trends.

Each prediction record includes:

Output Attribute	Description
Forecast Timestamp	Future reporting period
Forecast Value	Predicted adverse event report count
Lower Bound	Lower confidence interval
Upper Bound	Upper confidence interval

These predictions provide a forward-looking view of reporting activity and can be integrated into business dashboards for continuous monitoring.

	SERIES	TS	FORECAST	LOWER_BOUND	UPPER_BOUND
1	"MFG-BR-SAO-04"	2026-01-01 00:00:00.000	2.742211453	-0.3402565442	5.81391439
2	"MFG-BR-SAO-04"	2026-02-01 00:00:00.000	2.962261705	-0.03466463304	5.847303288
3	"MFG-BR-SAO-04"	2026-03-01 00:00:00.000	3.161016771	0.2579699267	6.076897804
4	"MFG-DE-BER-02"	2026-01-01 00:00:00.000	1.33104272	-2.291861387	4.953946826
5	"MFG-DE-BER-02"	2026-02-01 00:00:00.000	1.33104272	-2.291861405	4.953946844
6	"MFG-DE-BER-02"	2026-03-01 00:00:00.000	1.33104272	-2.291861423	4.953946862
7	"MFG-IN-HYD-03"	2026-01-01 00:00:00.000	0.7205355716	-1.146233235	2.753961108
8	"MFG-IN-HYD-03"	2026-02-01 00:00:00.000	0.7031122121	-1.410669068	2.748586792
9	"MFG-IN-HYD-03"	2026-03-01 00:00:00.000	0.6856888088	-1.542660829	2.741464366
10	"MFG-JP-OSA-05"	2026-01-01 00:00:00.000	1.667465461	-2.681789496	6.016720419
11	"MFG-JP-OSA-05"	2026-02-01 00:00:00.000	1.667465461	-2.681789518	6.016720441
12	"MFG-JP-OSA-05"	2026-03-01 00:00:00.000	1.667465461	-2.681789539	6.016720462
13	"MFG-US-ATL-01"	2026-01-01 00:00:00.000	0.9660050896	-0.5512086547	2.649787617
14	"MFG-US-ATL-01"	2026-02-01 00:00:00.000	0.9422790379	-0.7641804387	2.584968518
15	"MFG-US-ATL-01"	2026-03-01 00:00:00.000	0.9208490558	-0.8557718534	2.498961665

Figure 8.1 presents the forecast prediction output generated by Snowflake Cortex Forecasting.

8.6 Business Value of Forecasting

The forecasting capability enhances pharmacovigilance operations by enabling proactive analysis of reporting trends rather than relying solely on historical data.

Key business benefits include:

- Anticipating changes in adverse event reporting volumes.
- Supporting early review of increasing reporting trends.
- Assisting safety teams in planning investigations and resource allocation.
- Providing predictive insights alongside historical analytics.
- Enhancing decision-making through AI-assisted forecasting.

It is important to note that the forecasting model estimates future reporting volumes and **supports** safety monitoring. Final clinical assessment and regulatory decisions continue to rely on expert review.

9. Business Intelligence Dashboard

9.1 Dashboard Overview

The Business Intelligence layer serves as the final component of the Pharmacovigilance Intelligence Platform, transforming curated adverse event data into interactive visualizations that support operational monitoring and business decision-making.

The dashboard was developed using Microsoft Power BI and connected to the curated datasets stored in Snowflake. It provides a centralized interface for monitoring adverse event reporting patterns across multiple analytical dimensions, enabling users to identify trends, compare reporting activity, and explore forecasted reporting volumes through interactive visualizations.

By integrating historical reporting data with AI-generated forecasting results, the dashboard enables both retrospective analysis and predictive monitoring within a single reporting environment.

9.2 Dashboard Objectives

The dashboard was designed to provide business users with a comprehensive view of pharmacovigilance reporting while minimizing the effort required to analyze large volumes of adverse event data.

The primary objectives of the dashboard are to:

- Provide real-time visibility into adverse event reporting trends.
- Enable analysis by drug, manufacturing site, country, and reporting period.
- Present key performance indicators through intuitive visualizations.
- Support proactive decision-making using forecasted reporting trends.
- Deliver a user-friendly interface for interactive data exploration.

9.3 Dashboard Components

The dashboard consists of multiple analytical components, each designed to address a specific business requirement.

Dashboard Component	Purpose
KPI Cards	Display summary metrics such as total adverse event reports, number of drugs, manufacturing sites, and countries.
Trend Analysis	Visualizes adverse event reporting patterns over time.
Drug Analysis	Compares reporting volumes across different pharmaceutical products.
Manufacturing Site Analysis	Evaluates reporting activity across manufacturing facilities.
Geographic Analysis	Displays adverse event distribution by country.
Forecast Analysis	Compares historical reporting trends with forecasted values generated by Snowflake Cortex Forecasting.

9.4 Interactive Features

The dashboard incorporates interactive capabilities that allow users to perform detailed exploratory analysis without modifying the underlying data model.

Implemented features include:

- Dynamic filtering using slicers.
- Cross-filtering between visualizations.

- Drill-down into reporting periods.
- Interactive tooltips displaying detailed information.
- Responsive navigation across analytical views.

These features enable users to quickly investigate reporting trends and identify areas requiring further review.

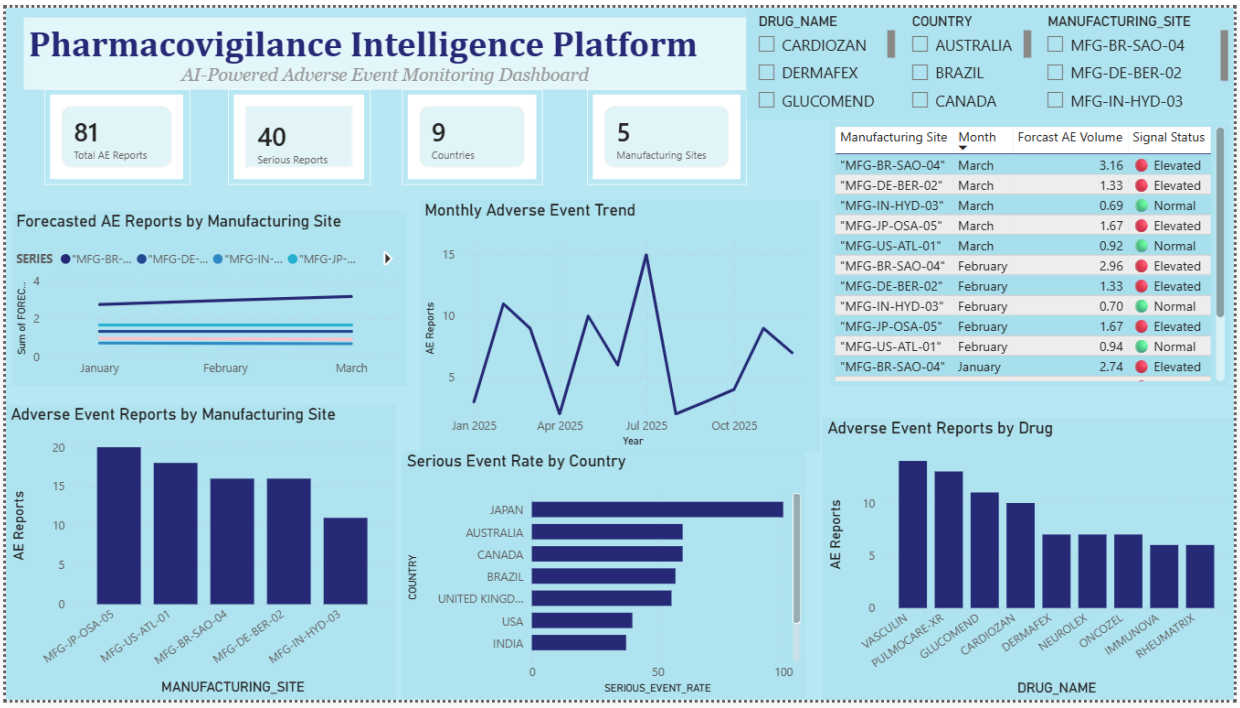


Figure 9.1 presents the complete Business Intelligence dashboard developed for the project.

10. Project Outcomes and Business Benefits

10.1 Project Outcomes

The Pharmacovigilance Intelligence Platform successfully transformed a manual adverse event reporting process into an automated, cloud-native data engineering solution. By integrating AWS, Dataiku DSS, Snowflake, Snowflake Cortex, and Power BI, the platform establishes a structured workflow that supports data ingestion, quality management, analytical modeling, forecasting, and business reporting.

The implemented solution achieved the following outcomes:

- Automated ingestion of raw adverse event reports into a secure cloud environment.

- Improved data quality through profiling, cleansing, standardization, validation, and deduplication.
- Development of a dimensional data model optimized for analytical reporting.
- Creation of reusable analytical views to support reporting and AI/ML workloads.
- Implementation of Snowflake Cortex Forecasting to estimate future adverse event reporting volumes.
- Delivery of an interactive Power BI dashboard for monitoring historical and forecasted reporting trends.

Together, these capabilities provide a scalable foundation for pharmacovigilance analytics while reducing manual effort in data preparation and reporting.

10.2 Project Deliverables

The project successfully delivered the following components.

Deliverable	Description	Status
AWS S3 Landing Zone	Secure storage for raw adverse event data	Completed
Snowflake Data Warehouse	RAW, STAGING, and CURATED layers	Completed
Dataiku ETL Pipeline	Profiling, cleansing, validation, and deduplication	Completed
Dimensional Data Model	Fact and dimension tables	Completed
Analytical Views	Reporting and Snowflake Cortex Forecast-ready datasets	Completed
Snowflake Cortex Forecasting	Future adverse event reporting predictions	Completed
Power BI Dashboard	Interactive reporting and visualization	Completed
Project Documentation	Technical design and implementation document	Completed

11. Future Enhancements

11.1 Overview

While the implemented Pharmacovigilance Intelligence Platform successfully achieves the defined project objectives, the solution has been designed with scalability and extensibility in mind. The cloud-native architecture provides a flexible foundation that can accommodate additional analytical capabilities, larger datasets, and evolving business requirements. Future enhancements can further improve automation, predictive analytics, and regulatory compliance within pharmacovigilance operations.

11.2 Real-Time Data Integration

The current implementation processes adverse event reports using batch-based CSV ingestion. Future versions of the platform can integrate with real-time data sources through APIs or streaming technologies, enabling continuous ingestion of adverse event reports as they are generated. This enhancement would reduce processing latency and support near real-time safety monitoring.

11.3 Advanced AI and Machine Learning

The project currently utilizes Snowflake Cortex Forecasting to predict future adverse event reporting volumes. Additional AI capabilities can be incorporated to strengthen pharmacovigilance analytics, including:

- Automated anomaly detection for identifying unusual reporting patterns.
- Classification models for adverse event prioritization.
- Predictive risk scoring based on historical safety data.
- Intelligent alert generation for potential safety signals.

These enhancements would further support proactive pharmacovigilance decision-making.

11.4 Natural Language Processing (NLP)

Many adverse event reports contain narrative descriptions that provide valuable clinical information. Future implementations can leverage Natural Language Processing (NLP) techniques to analyze textual narratives, extract medical entities, identify drug-event relationships, and support automated classification of adverse events.

11.5 Regulatory Reporting and Compliance

The platform can be extended to automate the preparation of standardized pharmacovigilance reports required by regulatory authorities. Integration with regulatory reporting frameworks would reduce manual effort, improve reporting consistency, and support compliance with evolving regulatory requirements.

11.6 Enterprise Scalability

The implemented architecture is designed to support future expansion across larger datasets, multiple data sources, and additional geographical regions. Future deployments may include automated scheduling, enhanced monitoring, role-based access control, and CI/CD pipelines to support enterprise-scale operations.

11.7 Future Enhancement Summary

The proposed enhancements demonstrate that the implemented platform provides a scalable foundation for future innovation. By incorporating real-time data processing, advanced AI capabilities, NLP-based analysis, and automated regulatory reporting, the solution can evolve into a comprehensive pharmacovigilance analytics platform capable of supporting modern healthcare and pharmaceutical organizations.

12. Conclusion

The Pharmacovigilance Intelligence Platform successfully demonstrates the implementation of a cloud-native data engineering solution for processing, managing, and analyzing adverse event reports. The project integrates AWS S3, Dataiku DSS, Snowflake, Snowflake Cortex, and Power BI into a unified workflow that automates the complete data lifecycle—from ingestion and quality management to analytical modeling, predictive forecasting, and interactive reporting.

Through the implementation of a layered architecture comprising RAW, STAGING, and CURATED data layers, the platform establishes a structured and governed approach to data management. Dataiku DSS ensures that adverse event data is profiled, cleansed, standardized, validated, and deduplicated before being transformed into analytics-ready datasets. Snowflake analytical views and Snowflake Cortex Forecast extend the platform beyond descriptive reporting by enabling predictive analysis of future adverse event reporting trends, while Power BI provides interactive dashboards that support business users in monitoring key safety indicators.

The project successfully achieved its primary objectives by automating data ingestion, improving data quality, implementing a dimensional data model, preparing reusable analytical datasets, integrating AI-assisted forecasting, and delivering interactive business intelligence dashboards. Together, these capabilities improve operational efficiency, strengthen data governance, and support informed decision-making within pharmacovigilance processes.

Overall, the implemented solution demonstrates how modern cloud-native data engineering technologies can be applied to transform traditional pharmacovigilance workflows into scalable, reliable, and analytics-driven platforms. The architecture provides a strong foundation for future enhancements and serves as a practical example of integrating data engineering, machine learning, and business intelligence to support proactive drug safety monitoring.